# What an Auxiliary Cue Must Carry: LiDAR Body Pose for Pedestrian Trajectory Forecasting

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***Abstract*—Gait and facing appear to encode pedestrian intent that a position history lacks, making body pose an attractive forecasting input. On the Waymo Open Motion Dataset (WOMD), with SMPL poses recovered from LiDAR, we find that whether it pays depends on the track it is added to, and on what the pose pipeline was itself given. The decisive comparison is correct pose against the same architecture fed *another pedestrian's* pose, which separates body information from an extra branch's capacity. On WOMD's tracks it returns $+0.04\%$ ($p$=0.93) at our largest scale, and the data bound the content effect to $\pm 1.7\%$. Those tracks are offboard auto-labels, refined with future frames. Replace the observed history with what a causal onboard tracker would report, and own pose beats a stranger's by $5.3\%$ ($p$=0.002) at $30$ cm of detection noise. Three quarters of that gain is the label returning through the body: the pipeline's motion-prior fit matches the gait to the offboard track, and with poses that never saw the track the gain is $1.3\%$ ($p$=0.10). What remains appears to grow with the LiDAR returns on the body, to $3.5\%$ ($p$=0.010) on pedestrians with at least $200$ against $0.3\%$ below $30$, in one of two stratifications we scored and not yet replicated. As a bearing, body facing is a noisier copy of the box heading the model already receives ($12.3^\circ$ against $3.9^\circ$). An oracle through the same branch converts the agent's own future bearing at every error level we trained at $25$k scenes, so the pathway is not the obstacle. Before integrating a modality, measure what it carries beyond the inputs the model already has, at the quality a vehicle will have, and check which labels its own pipeline consumed.**

***Index Terms*—pedestrian trajectory prediction, human pose estimation, LiDAR, autonomous driving, offboard labels**

## I. Introduction

Predicting where a pedestrian will go is a safety-critical function of an automated vehicle, and body pose is an obvious candidate input. A person's stance, gait phase and facing direction plausibly reveal intent before their position changes, and modern perception stacks already produce human mesh or keypoint estimates. Several recent systems condition trajectory forecasts on pose [1]–[4].

This paper reports that, on the labelled tracks of the Waymo Open Motion Dataset (WOMD), pose conditioning does not measurably improve pedestrian trajectory accuracy, and that the reason is the track it is added to. WOMD's boxes are offboard auto-labels, refined with the whole track, future frames included [5], [6]. Against that track the body restates what the model already knows. Against the history a causal onboard tracker would report, the same body appears to pay, and most of that gain is the offboard label returning through the pose pipeline, which fits the gait to the track. Poses that never saw the track appear to pay only where the sensor resolves the body. Both reasons can be checked quantitatively, and cheaply, before a modality is integrated at all.

We make four contributions.

**A controlled null on offboard tracks.** We measure pose conditioning against a shuffled-pose control: the same architecture receiving a different pedestrian's body motion. This isolates information content from the capacity cost of an extra branch, and it is the comparison a practitioner should demand before adopting any auxiliary modality. The control is null at three training scales and under two integration styles. It is also null across the three pose encodings we run it against on the corrected export (6D joint rotations, agent-frame root, and eight frame-invariant descriptors), with constant-pose as a capacity control. Two further encodings (3D joint positions and a root-orientation-free variant) and a GRU encoder predate the export fix of Sec. III and were run against no-pose rather than against a stranger's pose, so we neither count them as controlled nulls nor quote numbers from them. An oracle arm, which replaces the body with a signal encoding the agent's own future, improves minADE by $34.8\%$ through the same branch, so the null is a property of what pose adds to these tracks, not of the plumbing.

**An onboard-quality track, and a label leak.** We replace WOMD's observed history with what a causal tracker would report, per-frame detection noise filtered by a forward-only Kalman filter, and repeat the content test. Own pose now beats a stranger's by $5.3\%$ at 30 cm of detection noise. Three quarters of that is not body information. The pose pipeline ends in a motion-prior fit whose loss matches the gait to the box track, so the fitted body states the clean speed the tracker lost: body pose alone predicts it with $R^2$=0.94, against $0.44$ before the fit. Rerun on the pre-fit poses, the gain is $1.3\%$ and no longer significant. On well-resolved bodies it is $3.5\%$ and grows with LiDAR returns, a result from one stratification that awaits replication.

**What the pathway converts, and what it does not.** We corrupt that oracle with a controlled angular error to ask what kind of information the branch turns into accuracy. At 25k scenes every error level we trained converts, including one uniformly worse than an input the model already has, so the ladder does not set an accuracy a cue must reach. It separates a precision term with a $3.0^\circ$ half-life from a slowly decaying term that is future-side mode selection. Body facing supplies neither: it is a noisier copy of the box heading where pedestrians move, and reaches $41.2^\circ$ where they do not, which is exactly the case velocity cannot cover. We also show the same ladder run at 579 scenes reports a clean accuracy threshold that 25k scenes removes, so an underpowered ladder manufactures a floor where there is none.

**A measurement caveat.** Aggregate MPJPE cannot distinguish a pose estimator that tracks a joint imprecisely from one that ignores the input and emits its prior. Scoring each joint against an always-predict-the-mean-pose baseline, MPJPE is uninformative about which joints are actually recovered ($\rho = -0.11$ across joints, n.s.) and inverts the ordering for the head. Accepting a pose estimator into an AV stack on a single error number risks importing a prior.

We also report, as a cautionary note, that our first attempt at this study was invalidated by an export bug that silently substituted other pedestrians' bodies, a failure any pose pipeline keyed by object identity can reproduce.

## II. Related Work

Trajectory forecasting on WOMD is dominated by anchor- and query-based transformers. We build on MTR [7], [8], which is representative and strong. Human mesh recovery from LiDAR is maturing [9], with motion priors used to complete sparse observations [10]–[12]. Intent-focused work on pedestrian datasets [13], [14] targets crossing decisions rather than metric trajectories. Kress et al. [4] report pose-based gains for vulnerable road users from a research vehicle, at close range and with camera-derived pose. Methodologically our closest precedent is Makansi et al. [15], who find by feature attribution that trajectory predictors draw almost nothing from an input the field had assumed was central.

**Relation to prior positive results.** Pose conditioning has reported real gains, and our mechanism predicts where they come from rather than contradicting them. Social-Transmotion [1] lowers ADE on the synthetic JTA dataset from 0.99 to 0.89 with *ground-truth* 3D pose, and, in the harder case for us, from 0.40 to 0.36 on JRDB with *estimated* pose. Social-Pose [2] reports about 25% on JTA (1.20→0.90) and on JRDB. Salzmann et al. [3] report ≈10% from onboard-estimated keypoints at ranges up to 7 m, specifically for agents with little position history. A short history is a poor track and 7 m is a well-resolved body, the one condition under which our track-free poses appear to pay (Sec. IV-E).

Le et al. [16] reach the opposite conclusion on JRDB and, in doing so, say the most useful thing about ours: they localise the gain to *lower-body* 3D keypoints (13% ADE, a further 1–4% from derived biomechanical cues). Their cue is our weakest channel. Of the twelve limb joints in Table V the knees and ankles that carry gait sit near the bottom of the range (SKILL $+0.46$ to $+0.49$, against $+0.63$ and $+0.64$ for the elbows), and what leg motion we do recover is damped: predicted knee and ankle spread is 84–91% of ground truth, where elbows and wrists reach 96–99%. A panoramic camera at a few metres resolves a swinging leg. A few dozen LiDAR returns at 40 m recover its mean position and lose much of its amplitude. So the two results do not conflict: theirs is what this cue is worth when the limbs are resolved and the predictor has no heading input, ours what it is worth when the trajectory already carries the body's direction and speed. Degrade our track to a causal tracker's and a gain appears on the best-resolved bodies, pending replication (Sec. IV-E). On the offboard track sparsity does not account for the difference, since the content test is null even on those agents (Sec. IV-D).

The most informative comparison is within Social-Pose itself. Alongside JTA it evaluates on Pedestrians and Cyclists in Road Traffic, the one vehicle-captured dataset there, and, with *ground-truth* 3D pose, ADE moves from 0.58 to 0.57 m, by 0.01 m or less for all five architectures tested, against the ≈25% the same encoder delivers on JTA. The authors attribute this to the shorter horizon and the absence of interacting neighbours. Our measurements suggest an additional reason that travels with the sensor rather than the horizon. Either way, the closest published setting to ours already reports approximately our result.

Two properties separate the remaining positive settings from an AV stack, and both follow from the mechanism in Sec. V. *Range and sensor:* JRDB is an indoor and campus robot dataset in which pedestrians are metres away and pose is camera-derived, whereas WOMD pedestrians are observed from 2 to 75 m and often return a few dozen LiDAR points. *The baseline:* those predictors consume positions, not a box heading, so body facing is new information there and partly duplicated here. On controls, none reports a shuffled-pose arm, though the confound differs by design. Social-Transmotion masks modalities within one network, so no extra parameters are involved there, whereas a dedicated pose branch pays the capacity cost we measure (≈1.6% minADE), the same order as the smaller reported gains.

## III. Setup

### A. Data and a data-integrity failure

We use WOMD [5] with SMPL [17] poses recovered from LiDAR by the Waymo-3DSkelMo pipeline [18], which fits per-frame bodies with LiDAR-HMR [9] and completes them using AMASS-trained motion priors [10]–[12]. Poses cover the 11 observed frames, and the prediction horizon is 8 s at 10 Hz. In every arm the model sees the map and the pedestrians that carry poses. Vehicles, cyclists and the ego vehicle are not in its input. The branch is not starved: every predicted agent carries at least one valid pose frame, and 86.7% of observed frames carry one, so sparsity of pose is not an explanation for anything that follows.

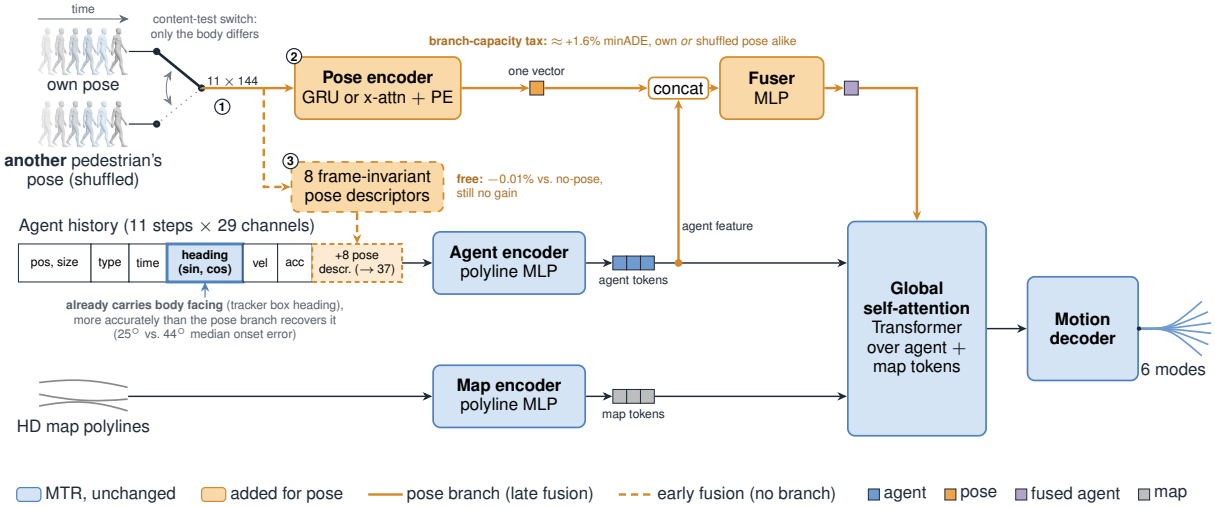



Fig. 1. The two arms differ only in *whose* body enters the model. The content-test switch (1) feeds the predictor either the pedestrian's own SMPL sequence or another pedestrian's, holding parameters, optimiser and input dimensionality fixed. Everything downstream is shared. Blue: MTR, unchanged. Orange: added for pose. The agent-history strip shows that the labelled box heading is *already* an input. Solid orange (2): the pose branch (late fusion), which costs ≈1.6% minADE whether it carries the right body or the wrong one. Dashed (3): early fusion, which removes that cost, still yields no benefit from pose, and converts an uncorrupted oracle on every seed at 25k scenes (Sec. V-C).

One property of that pipeline bears on Sec. V: when a pedestrian is moving and the raw LiDAR estimate of root rotation deviates from the past-trajectory direction by more than 90°, the pipeline snaps the root to the direction of travel. Body facing is therefore partly trajectory-derived for moving pedestrians. This reinforces the redundancy we report rather than weakening it, and it does not reach the stationary-onset analysis, whose events are all below 0.3 m/s. A second property bears on Sec. IV-E: the motion-prior stage fits each sequence against the box track, so the completed body is not independent of that track. We call the per-frame LiDAR-HMR output that precedes it the *pre-fit* pose.

Our first pass at this study was invalid. The export keyed pose clips by a scene-*local* object identifier, so clips collided across scenes and nearly every training sequence carried another pedestrian's body. A per-scene re-export (299,220 scenes, 1.81M sequences) cut the median error between root heading and travel direction from 41.5° to 11.4° and raised the gait–speed correlation from $r$=0.26 to 0.58. We report it because the failure is silent (training converges, losses look normal) and identity-keyed joins are common in pose pipelines. Every number here is from the corrected export.

### B. Arms and the content test

Three arms share one architecture (Fig. 1). *No-pose* receives trajectories and map. *Pose* adds the pedestrian's own SMPL sequence, 144 joint-rotation values per frame with no translation. *Shuffled* receives another pedestrian's sequence, assigned once over the whole split by a fixed derangement, with identical parameter count, optimiser and input dimensionality. The root orientation enters in world coordinates while the trajectories are agent-centric, so this arm does not state facing relative to the model's frame. The agent-frame-root encoding does, and its content test is null too (+0.68%, $p$=0.49, $n$=3, MDE 4.3 pp). A fourth arm, *oracle*, replaces the SMPL features with a six-channel encoding of the agent's own future displacement at 3 and 8 s, leaving the branch and its masks untouched. It shows the pathway can carry information at all (Sec. IV-C). Every arm with a pose branch, the stranger and the oracle included, is also trained to predict the future pose sequence it was given, under an auxiliary loss (a geodesic rotation term and a mixture likelihood term, weight 0.1 each). The no-pose arm has no such loss.

Pose versus no-pose confounds information with branch capacity and the auxiliary loss, while pose versus shuffled holds both fixed and isolates content. We call the latter the *content test* and treat it as decisive. Relative differences are $(A - B)/B$, so negative favours $A$.

Inference is over *seeds*, which is the unit the treatment is applied to: every comparison between trained arms is a paired $t$-test on per-seed differences. The two exceptions are flagged where they occur: per-pedestrian subpopulation tests and the per-joint intervals of Table V, which resample pedestrians. We also ran a hierarchical bootstrap that resamples seeds and then pedestrians. It agrees on the nulls and undercovers at three to five seeds, so it never replaces the paired test. We do not quote a percentile bootstrap $p$-value, because with five seeds whose differences share a sign it returns the resolution floor $1/N_{\text{boot}}$ rather than a calibrated tail probability, and a pedestrian-only bootstrap ignores between-seed variance entirely.

### C. Protocol

Every arm trains for 30 epochs with AdamW (two scenes per GPU with at most 24 predicted pedestrians each, weight decay 0.01, gradient-norm clipping 1.0) under a step schedule that halves the learning rate at epochs 10, 20 and 25, and

TABLE I

CONTENT TEST ACROSS TRAINING SCALE, VALIDATION FIXED. THE LEARNING RATE IS SELECTED PER SCALE BY A GRID RUN SEPARATELY FOR EVERY ARM, AND ALL ARMS PEAK AT THE SAME VALUE (SEC. III-C). PERCENTAGES ARE DIFFERENCES OF ARM MEANS, AND $p$-VALUES ARE PAIRED $t$-TESTS OVER PER-SEED DIFFERENCES. NEGATIVE FAVOURS POSE.

| Training scenes | 579 | 5k | 25k |
|---|---|---|---|
| Pedestrians | 6,348 | 31,302 | 142,807 |
| Seeds | 5 | 3 | 5 |
| *minADE* (m) | | | |
| No-pose | 0.4599 | 0.4176 | 0.3716 |
| Pose | 0.4620 | 0.4138 | 0.3778 |
| Shuffled | 0.4589 | 0.4181 | 0.3777 |
| Pose vs. no-pose | $+0.45\%$ | $-0.90\%$ | $+1.66\%^{*}$ |
| **Content test** | $+0.66\%$ | $-1.02\%$ | $+0.04\%$ |
| $p$ (content) | 0.52 | 0.56 | 0.93 |
| *minFDE* (m) | | | |
| No-pose | 0.9922 | 0.8992 | 0.8049 |
| Pose | 0.9824 | 0.8875 | 0.8131 |
| Shuffled | 0.9780 | 0.8900 | 0.8104 |
| Pose vs. no-pose | $-0.99\%$ | $-1.30\%$ | $+1.02\%^{\dagger}$ |
| **Content test** | $+0.45\%$ | $-0.28\%$ | $+0.33\%$ |
| $p$ (content) | 0.63 | 0.83 | 0.61 |

$^{*}t(4)=10.3$, $p=5\times10^{-4}$. The shuffled arm pays the same cost, $+1.63\%$, $t(4)=2.8$, $p=0.048$.
$^{\dagger}t(4)=6.1$, $p=0.004$.

TABLE II

CLEAN CORE AT 579 SCENES, $n=5$ SEEDS, BOTH REGIMES. THE SHUFFLED ARM HAS THE LOWEST MEAN IN BOTH: ANOTHER PEDESTRIAN'S BODY IS NOMINALLY THE BEST INPUT.

| Arm | Per-seed minADE | Mean ± sd | Content |
|---|---|---|---|
| *From scratch* | | | |
| No-pose | .458 .443 .453 .495 .450 | $0.4599 \pm 0.0206$ | — |
| Shuffled | .454 .467 .451 .455 .467 | $0.4589 \pm 0.0075$ | — |
| Pose | .471 .456 .453 .457 .474 | $0.4620 \pm 0.0094$ | $+0.66\%$ |
| *Fine-tuned on a full-WOMD backbone* | | | |
| No-pose | .372 .366 .367 .373 .368 | $0.3692 \pm 0.0031$ | — |
| Shuffled | .364 .363 .364 .374 .373 | $0.3676 \pm 0.0055$ | — |
| Pose | .364 .372 .371 .374 .373 | $0.3708 \pm 0.0040$ | $+0.88\%$ |

the decoder predicts $K=6$ modes. The learning rate is chosen per scale from a grid run separately for each arm: $1\times10^{-4}$ at 579 scenes, $5\times10^{-5}$ at 5k and $2.5\times10^{-5}$ at 25k, where every arm peaks at the same interior optimum. Validation is fixed at 5,171 scenes. Of their 17,218 predicted pedestrians, the 16,486 with at least one valid future step are scored. Headline numbers are taken at each run's best validation epoch over a fixed evaluation schedule, and Sec. IV-B reports what changes under a fixed-epoch rule. Official WOMD mAP and gated miss rate need the full retained prediction set, which our per-epoch evaluation does not store, so we report minADE, minFDE and miss rate at 2 m. minADE averages over a pedestrian's valid future steps and minFDE is taken at the last valid one, and 36% of the scored pedestrians have a full 8 s future. Code, configs, per-seed metrics and the analysis scripts are released with the paper.

## IV. RESULTS

### A. The null on offboard tracks, and its bound

Table I trains all three arms at three scales. Data alone improves the baseline by 19.2%, while the content test is null at every scale, in minADE and in minFDE alike, and in miss rate at 2 m, where the content test is $+2.2\%$ at 25k ($p$=0.21) and $+4.3\%$ fine-tuned ($p$=0.19): null, and with its sign against pose on all three metrics. The scales are not equally sensitive, and we state what each one could have seen: at 80% power the minimum detectable content effect is 3.5% at 579 scenes ($n$=5), 7.6% at 5k ($n$=3) and 2.2% at 25k ($n$=5). The 5k scale is therefore weak evidence on its own and we do not count it as an independent null. The weight falls on 25k, which is both the largest scale and the most sensitive. At 25k the paired seed-level interval on the content difference is $[-1.59\%, +1.70\%]$, so at the largest scale we train the data exclude a content effect beyond roughly 1.7% in either direction, so the largest benefit the interval allows is no larger than the 1.6% cost of the branch that would carry it. The hierarchical bootstrap gives a visibly narrower interval ($[-0.0040, +0.0047]$ m, $\approx\pm1.2\%$). With five clusters it undercovers, so we quote the wider seed-level bound throughout.

After pretraining a trajectory-only backbone on full WOMD and fine-tuning ($n$=5), the content test is $+0.88\%$ ($p$=0.18), again with its sign against pose (Table II).

Seed count decides what Table II appears to say. On the first three seeds the same runs read as *pose harms* from scratch ($+1.85\%$) and *shuffled is best* after fine-tuning ($-1.28\%$). At $n$=5 both shrink three- to fourfold. Both three-seed effects are significant under the hierarchical bootstrap ($p$=0.037, $p$=0.016) and neither is under a paired $t$ over the same seeds ($p$=0.18, $p$=0.12): three seeds do not resolve a 1% effect here, and a bootstrap over three clusters will not say so.

### B. Two effects that are real, and are not pose

**A branch-capacity tax.** At 25k both pose arms are worse than no-pose ($+1.66\%$ at $p=5\times10^{-4}$ and $+1.63\%$ at $p$=0.048) and indistinguishable from each other. An extra input branch costs $\sim$1.6% whether it carries the right body or the wrong one. Both arms also train with the auxiliary pose loss, which we did not separate from the branch at this scale.

This is an operating-point statement, and the operating point matters: it is measured at each run's best validation epoch. The sign of a pose-versus-no-pose comparison is not stable under a fixed late-epoch rule: at epoch 30 pose is 8.0% better with $n$=3, where the no-pose arm had degraded further, and 0.9% worse with $n$=8, both n.s. and neither resolved at these seed counts. What is stable is that the comparison moves with the stopping rule, so we never compare architecturally different arms at a fixed late epoch. The *content* test is not significant under either rule, though it is noisier at epoch 30 ($-5.9\%$, $-4.7\%$ and $-1.7\%$ at the three scales, $p$=0.16, 0.07 and 0.60). Its two arms have identical input dimensionality, so this asymmetry does not arise. Replacing the 144-d branch with eight frame-invariant descriptors fused into the existing

agent channels removes the cost ($-0.01\%$ versus no-pose at $p$=0.99, and $-1.67\%$ versus the branch at $p$=0.12, $n$=3) without creating a benefit: pose can be carried for free, and still does not pay. That freedom is real, and Sec. V-C is what establishes it: fed an uncorrupted oracle at 25k scenes this pathway converts it on every seed, so a channel it costs nothing to carry is still a channel it reads. We quote the seed-level $p$ here for consistency with the rest of the paper. The hierarchical bootstrap makes the branch contrast look decisive ($p$=0.006) at three seeds, which is the undercoverage Sec. III warns against.

**Tuning dominates the treatment.** At 5k the learning rate inherited from the smaller scale made pose look catastrophic ($+37\%$). With per-arm tuning it reads $-0.90\%$, n.s. Across the 25k grid the learning rate moves minADE by up to $16.7\%$ against a content signal of $\sim 0.05\%$: a pose-versus-baseline comparison that does not tune each arm separately is a claim about the learning rate.

### C. The branch is not inert

A null measured through one pathway invites the obvious objection that the pathway is dead. The shuffled control does not dispose of it: it shows the model cannot distinguish its own body from a stranger's, which is equally consistent with a branch that transmits nothing at all. The branch-capacity tax shows the branch perturbs the output, not that it carries signal.

We therefore ran the positive control. The *oracle* arm replaces the SMPL features with a six-channel encoding of the agent's own future displacement at 3 and 8 s, which is genuinely predictive by construction. It keeps the branch, its parameter count, its input dimensionality and its per-frame validity masks exactly as the pose arm has them, so the oracle is visible only where a real pose would have been.

Over three seeds at 579 scenes the oracle arm reaches $0.2945$ against no-pose's $0.4514$ on the same seeds, a change of $-34.8\%$ ($t(2)=-81.4$, $p=1.5\times10^{-4}$), and $-35.6\%$ against the shuffled arm. The oracle is of course not a usable model, since it reads the future. Its only job is to calibrate the pathway. It does so emphatically: the branch converts information into accuracy at roughly twenty times the magnitude of the bound our content test places on pose. The null is therefore a statement about what SMPL pose contains, not about whether the architecture can use what it is given.

### D. Where a benefit could still hide

On an eight-seed replication of the content test at 30 epochs, scored on a held-out half of validation (about 8,040 agents), it is null at every horizon and least favourable at the shortest ($+0.48\%$ at 2 s, $-0.04\%$ at 8 s). It is null in all 20 tests across quartiles and top deciles of constant-velocity residual, heading variation, speed range and peak acceleration, on the 6,764 of those agents for which the four are defined. It is also null for standing-then-moving pedestrians, the subpopulation built to favour it ($n$=1,809 pedestrians, $+0.38\%$, epoch-30 checkpoints on an expanded validation split, $p$=0.80 by a pedestrian-resampling bootstrap). That is the one place we quote a bootstrap, because this subpopulation is defined per pedestrian.

TABLE III

DEGRADED INPUTS AT 25K SCENES, $n$=3 SEEDS PER ARM FOR THE FRAME ROTATION AND $n$=5 FOR TRACK NOISE. EACH BLOCK PERTURBS ONE INPUT OF THE BASELINE AND REPEATS THE CONTENT TEST. LOSS IS THE NO-POSE BASELINE AGAINST ITS CLEAN CELL ON THE SEEDS BOTH SHARE, AND NEGATIVE FAVOURS POSE. THE FRAME ROTATION LEAVES THE CLEAN HEADING READABLE, SO IT IS NOT AN ONBOARD-QUALITY HEADING (SEC. V-B). THE FITTED POSES SAW THE OFFBOARD TRACK, SO THEIR TRACK-NOISE ROWS OVERSTATE WHAT THE BODY ADDS. THE LAST ROWS REPEAT THE 30 CM LEVEL ON PRE-FIT POSES, OVERALL AND BY MEAN LIDAR RETURNS.

| Degraded input | Loss | Content | $p$ | vs. no-pose |
|---|---|---|---|---|
| None | — | $+0.41\%$ | 0.60 | $+1.97\%$ |
| *Agent frame rotated by a fixed error $\mathcal{N}(0,\sigma^2)$* | | | | |
| $\sigma$=15° | $+5.8\%$ | $-0.14\%$ | 0.90 | $-0.39\%$ |
| $\sigma$=30° | $+12.6\%$ | $-0.11\%$ | 0.93 | $+1.34\%$ |
| *Track, causal tracker on detection noise $s$, fitted poses* | | | | |
| $s$=10 cm | $+14.1\%$ | $-2.08\%$ | 0.005 | $-1.21\%$ |
| $s$=30 cm | $+32.3\%$ | $-5.32\%$ | 0.002 | $-4.06\%$ |
| *Track at $s$=30 cm, pre-fit poses* | | | | |
| All pedestrians | $+32.3\%$ | $-1.34\%$ | 0.10 | $+0.36\%$ |
| $<30$ returns | — | $-0.34\%$ | 0.72 | $+1.82\%$ |
| $\geq 200$ returns | — | $-3.47\%$ | 0.010 | $-1.99\%$ |

It is also null where the body is best measured, which is where a pose-quality explanation predicts an effect should appear. Stratifying the 25k content test by LiDAR return count gives $-0.37\%$ ($[-1.87, +1.14]$) on the 2,372 agents with at least 200 returns and $-0.08\%$ on the 10,902 with at least 30, and stratifying by range gives $-0.14\%$ ($[-1.61, +1.33]$) inside 10 m. No stratum shows an effect, at $n$=5 throughout. Whatever the null is, it is not only sparsity at range.

Across the more than forty content tests this study reports outside the onboard-track ladder of Sec. IV-E, two reach $p<0.05$, and both favour the stranger. In the stationary subpopulation of the heading ablation, correct pose is $+1.8\%$ *worse* than a stranger's ($n$=8, $p$=0.011), and in the stationary subset of the frame-rotation control at $\sigma$=30° it is $+1.3\%$ worse ($n$=3, $p$=0.049, Sec. V-B). At that multiplicity one or two such results are expected by chance, and their direction is against a pose mechanism rather than for one. We report them so that the released metrics contain no surprise. The tests of Sec. IV-E that favour pose are a different kind of result. They grow with the noise and agree in sign on every seed, and that section shows that most of what they measure is a label, not a body.

### E. An onboard-quality track, and a label leak

Everything above is measured on WOMD's tracks, and those are offboard auto-labels. An offline network refines every box with the whole track, future frames included [5], [6], and the velocity the baseline receives is exactly the backward difference of those refined positions at all 954,751 pedestrian time steps we checked. The observed history is as clean as the labeller could make it. A deployed stack has a causal tracker instead, so we simulate one. Each observed position receives detection noise $\mathcal{N}(0, s^2)$ per axis, drawn once rather

than per epoch, and a forward-only constant-velocity Kalman filter, started at rest (acceleration noise 1 m/s$^2$, initial velocity uncertainty 1.5 m/s), turns the detections into the positions and velocities the model sees. The box heading and the future targets are untouched. At prediction time, measured on 675 pedestrians, the tracker is off by 7.5 cm and 0.23 m/s in median at $s$=10 cm, and by 21 cm and 0.46 m/s at 30 cm, against a median pedestrian speed of 1.1 m/s.

With the pipeline's fitted poses the null breaks (Table III). At 30 cm own pose beats a stranger's by 5.32% ($p$=0.002, 95% CI $[-7.3, -3.4]$) and beats no pose by 4.06% ($p$=0.004), over five seeds. All five agree in sign ($-2.7$ to $-6.5\%$), and the mean holds at every fixed late epoch ($-5.2$ to $-5.8\%$ at epochs 26, 28 and 30). At 10 cm it is $-2.08\%$ against shuffled ($p$=0.005) and $-1.21\%$ against no pose ($p$=0.07). It sits where gait would put it. On the 8,757 pedestrians walking at 0.8 to 1.6 m/s it is $-3.3\%$ ($p$=0.008) at 10 cm and $-6.8\%$ ($p$=0.0002) at 30 cm, while on the 3,336 standing at prediction time it is significant at neither. It read to us as a stride restoring the speed the tracker lost.

**Most of it is the label.** The fitted poses are not independent of the track. The motion-prior stage [11] is a latent optimisation whose loss predicts the root translation from the local body motion and matches it to the box track, so the fit pulls the gait toward a stride that reproduces the offboard displacement. The model input holds joint rotations only and covers only the 11 observed frames, so auditing the input does not reveal it. A check without training does. On the 11,588 validation pedestrians with a pose on every observed frame, gradient-boosted trees predict the clean one-second speed from body pose alone, root excluded, with $R^2$=0.94 from the fitted pose and 0.44 from the pre-fit one. On moving pedestrians with fewer than 30 LiDAR returns it is 0.78 against less than zero: the fitted body knows a speed the sensor could not have shown it. Added to a regressor on the 30 cm tracker's own state, the fitted pose removes 71% of the squared speed error that regressor leaves, the pre-fit pose about a fifth and a stranger's 1%. At 10 cm the shares are 21% and under 2%.

We therefore reran both pose arms at 30 cm on the pre-fit poses, which see only the points inside the box. Files, agents, tracks and the stranger assigned to each pedestrian are unchanged, and the two shuffled arms agree (+0.40%, $p$=0.54). Own pose now beats a stranger's by 1.34% ($p$=0.10, CI $[-3.1, +0.4]$) and does not beat no pose (+0.36%, $p$=0.54). At fixed epochs 26, 28 and 30 it reads $-1.4$, $-2.6$ and $-1.5\%$, significant at the second only ($p$=0.02). The fit accounts for 3.98 of the 5.32 points ($p$=0.0001, between 3.5 and 4.9 on every seed), and on walkers $-6.8\%$ becomes $-0.8\%$ ($p$=0.36).

**What survives appears to follow the sensor.** Split by mean LiDAR returns over the observed second, the pre-fit content effect is $-0.3\%$ below 30 returns, $-1.0\%$ and $-1.4\%$ in the two middle strata, and $-3.47\%$ ($p$=0.010, CI $[-5.6, -1.4]$) on the 2,365 pedestrians with at least 200, where it also beats no pose ($-1.99\%$, $p$=0.041). The top stratum is better than the bottom on every seed, by 3.1 points ($p$=0.004), and at every fixed late epoch it reads $-3.2$ to $-4.6\%$. The bottom stratum is less settled. At those epochs it reads $-1.4$ to $-2.1\%$ (n.s.), so the gap between the two narrows to 1.7 to 2.5 points. The stranger arm is also 0.9% worse on pre-fit than on fitted poses in the top stratum ($p$=0.015), and against the fitted stranger the effect is $-2.6\%$ ($p$=0.03). The fitted poses show no gradient ($-5.3$, $-5.4$, $-4.1$ and $-6.0\%$, $p$=0.41), as a leak that does not depend on the sensor would not. On the offboard track the same stratum is null (Sec. IV-D).

Three caveats bound this. The tracker is simulated, with Gaussian detection noise rather than a detector's correlated, occlusion-driven errors. The pre-fit poses are per-frame and unsmoothed, so a prior fitted to the body alone might recover more, though on the 347 frames with labelled keypoints the fitted poses are no more accurate than the pre-fit ones (92.7 against 87.6 mm MPJPE). And returns are one of two stratifications we scored at this noise level, four strata each, so $p$=0.010 is uncorrected over eight strata and the gradient needs replication. What the two runs establish is that the same body adds nothing to an offboard track and that a pose fitted to a labelled track inherits the label. That it adds to a causal track where the sensor resolves it rests on one stratification.

## V. Why Pose Does Not Pay on Offboard Tracks

### A. What the oracle ladder measures

The content test bounds what pose is worth, and the oracle of Sec. IV-C shows the branch can carry information. Neither says what *kind* of information this pathway turns into accuracy. We measured that by degrading the oracle.

Each level of Table IV feeds the branch the same direction-only oracle, the bearing of the agent's own future displacement at 3 and 8 s, corrupted by a fixed angular error. Each is labelled by the median bearing error actually delivered: $0.6745\sigma$ by construction, since the corruption is a fixed per-agent rotation, and measured to within 0.7° of it on 11,328 scored agents. Displacement magnitude is withheld, so the channel carries direction alone. The error is drawn once per pedestrian, not per epoch: a real estimator's error is a property of the observation, and resampling it would let training average it away.

At 25k scenes every level converts, and the gain stays large: $-34.50\%$ with an exact bearing, $-19.77\%$ at 5.0°, $-15.68\%$ at 10.0° and $-13.57\%$ at 15.1°, all at $p \leq 0.0016$ against minimum detectable effects of 1.4–2.9 pp. The decay is not exponential: one exponential fitted to the four levels leaves residuals (measured minus fitted) of $-1.5$, $+3.2$, $+0.3$ and $-2.5$ pp, a systematic pattern in $\theta$ against those MDEs rather than scatter about the curve. The curve decelerates, and two terms fit it to 0.4 pp:

$$\Delta minADE(\theta) \;\approx\; -\big[\,13.1\% + 21.4\% \cdot e^{-\theta/4.4^\circ}\,\big], \qquad (1)$$

a precision term with a half-life of 3.0° over a term taken as flat across the fitted span, an assumption the half-life depends on. The seeds agree closely: 12.3–14.2% persistent, amplitudes 20.8–21.8%, half-lives 2.8–3.4°. That term is slow, not permanent, and two levels held out of the fit measure it. By 24.9° the precision term is below 0.1%, so that level reads the slow term almost alone, and it still converts: $-10.39\%$

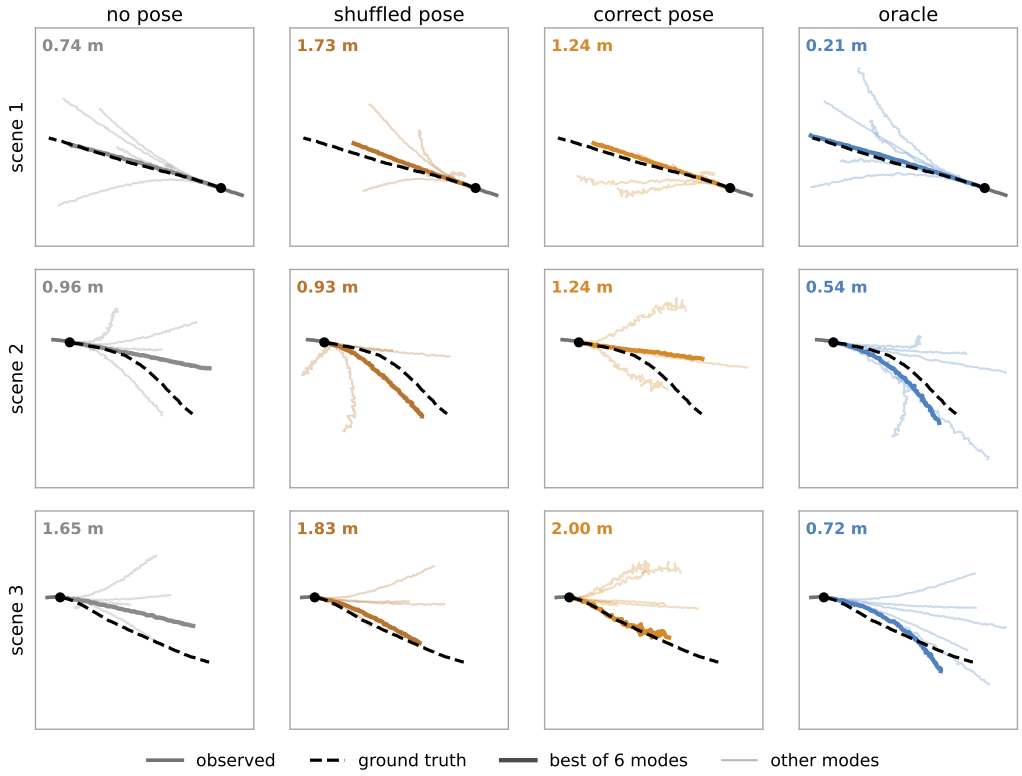



Fig. 2. Qualitative comparison of the four arms on the same three agents (seed 101, epoch 30, the epoch for which full trajectory dumps were retained). Scenes are picked by percentile of the *no-pose* arm's minADE (60th, 76th, 92nd) among agents with a full 8 s future and more than 3 m of travel. Selection is on the baseline alone, never on the treatment. Thin lines are the six predicted modes, thick the best of them, dashed the ground truth, grey the observed past. Fed the agent's own future bearing, the oracle bends its modes onto the true path through the *same* branch that carries pose, while correct pose is not consistently distinguishable from a stranger's pose or from no pose. Per-panel values are single-seed illustrations: across seeds the oracle's advantage is stable ($-34$ to $-40\%$), whereas the content difference at this fixed epoch is not ($-16.2$, $-8.0$, $-0.2\%$ for seeds 101/202/303) and is null at the best-epoch operating point used for inference (Table I).

($p$=0.003, CI $[-12.9, -7.9]$) against the $-13.2\%$ Eq. (1) extrapolates (Table IV). Four fifths of the term survive a 25° median error. At 44.4° the gain is $-6.92\%$ ($p$=0.013, CI $[-10.3, -3.5]$), so about half of it is left. It must vanish eventually, since a 60° median error is close to uniform on the circle.

**The angular axis is not an acceptance threshold.** It is tempting to read $\theta$ as the accuracy a cue must reach: feed the branch a channel whose bearing error is $\theta$, expect the gain at level $\theta$. The ladder refutes that reading itself. The baseline already receives the labelled box heading, as a per-timestep sine–cosine pair for every agent (Fig. 1), and that heading estimates the *same* future displacement bearing at 3.9° in median, better than the 15.1° channel at every percentile (9.3° against 25.8° at the third quartile, 20.1° against 36.3° at the ninetieth). A channel uniformly less accurate than an input the model already has still buys $-13.6\%$. Nor does the branch pay for merely being occupied: an extra branch carrying uninformative content *costs* $\sim$1.6% (Sec. IV-B), and through it the pedestrian's own pose differs from a stranger's by $+0.04\%$ (Table I). The 34.5% is content, and the only content this channel has is the realised future.

**The surviving term is future-side mode selection.** Eq. (1) splits the gain, and the two terms behave differently in ways that identify them. Per-agent records at each run's own best epoch, paired between arms on 16,486 agents and averaged over seeds, put the persistent term at the end of the trajectory: refitting the decomposition at each horizon leaves the precision half-life at 2.9–3.9° throughout, while the persistent share of the gain rises from 21% of the 2 s error reduction to 39% at 8 s and 53% at the endpoint. The selection reading predicts exactly that, and it predicts more: corrupting the bearing should barely affect *which* of the six modes is right. It does not. The gap between the top-ranked mode and the best of six, which is what the forecaster loses by choosing wrongly, falls from 0.67 m to 0.26 m with an exact oracle, and still to 0.29 m at 15.1°: the corrupted channel closes 92% of the perfect oracle's selection gap while recovering 39% of its minADE. The absolute gain is also largest in the quartile where the baseline's own modes disagree most at the endpoint (0.150 m against 0.111–0.130 m elsewhere, at $\theta$=0). A coarse future bearing says which way the agent went, and only a precise one refines where along that way, and that refinement is what 3° of error destroys.

minADE takes the best of six modes whatever their rank, so selection reaches it only through the decoder's scoring step, which keeps six of 64 intention queries by non-maximum suppression on predicted likelihood, with a 2.5 m endpoint radius: a better ranking changes which six survive. That is also where a pose effect could hide from our headline metric, so we scored the ranking columns directly on the 25k arms. They are null as well: against a stranger's pose, top-1 ADE is $-2.07\%$ ($p$=0.11, MDE 3.7 pp), best-mode rank $-0.93\%$ ($p$=0.22) and endpoint mode spread $+0.17\%$ ($p$=0.53), over the same 16,486 agents.

The ladder is therefore a positive control and an upper bound on what any direction channel through this branch could buy *if it knew the future*, and (1) says how much of that bound is precision rather than selection. An observation-time cue buys selection only to the extent that it predicts the realised mode beyond what the model's existing inputs already predict. The oracle's advantage is that it does not predict the future mode but reads it.

**A null at low power looks like a threshold.** We first ran this ladder at 579 scenes, where it reads as a clean detection floor: significant at 10.0° ($-6.0\%$, $p$=0.04), indistinguishable from zero at 15.1° ($-1.4\%$, $p$=0.47), flat above. At 25k that second level is $-13.6\%$. The 579 ladder could never have seen it: scaling its own $\theta$=0 amplitude ($-15.8\%$) by the 25k decay predicts $\approx$6.2% there, against that level's MDE of 6.7 pp. Scaled the same way, its 24.9° level should have read $\approx$4.7%, below that level's MDE of 6.6 pp, so its flat reading above 15.1° was blindness too. Its apparent floor was the angle at which an experiment of three to five seeds went blind, and a threshold read off the first non-significant level is a statement about power wearing the clothes of a property of the task. We report both ladders in Table IV for that reason.

TABLE IV

THE ORACLE LADDER. A DIRECTION-ONLY ORACLE CORRUPTED BY A FIXED PER-AGENT ANGULAR ERROR, EACH LEVEL LABELLED BY THE MEDIAN BEARING ERROR ACTUALLY DELIVERED. TESTS ARE PAIRED $t$ AGAINST EACH SEED'S OWN NO-POSE RUN, AND NEGATIVE FAVOURS THE ORACLE. MDE IS THE MINIMUM EFFECT A LEVEL COULD DETECT AT 80% POWER GIVEN ITS REALISED SPREAD. AT 25K SCENES EVERY LEVEL CONVERTS, OUT TO 44.4°. THE LAST TWO LEVELS ARE HELD OUT OF EQ. (1). EVEN 15.1° IS WORSE AT EVERY PERCENTILE THAN THE BOX HEADING THE MODEL ALREADY RECEIVES. THE LADDER THEREFORE BOUNDS WHAT A DIRECTION CHANNEL COULD BUY RATHER THAN SETTING AN ACCURACY A CUE MUST REACH (SEC. V-A). THE SAME LADDER AT 579 SCENES REPORTS A FLOOR AT 15.1° THAT IS ITS OWN MDE: PROPORTIONALLY TO ITS $\theta$=0 AMPLITUDE IT SHOULD HAVE READ $\approx$6.2% THERE, AND IT COULD RESOLVE 6.7.

| Bearing error | $n$ | ΔminADE | $p$ | 95% CI | MDE |
|---|---|---|---|---|---|
| *25k scenes* | | | | | |
| 0.0° | 3 | −34.50% | 0.0001 | [−35.6, −33.4] | 1.4 |
| 5.0° | 3 | −19.77% | 0.0007 | [−22.1, −17.5] | 2.9 |
| 10.0° | 3 | −15.68% | 0.0009 | [−17.7, −13.6] | 2.6 |
| 15.1° | 3 | −13.57% | 0.0016 | [−15.9, −11.2] | 2.9 |
| 24.9° | 3 | −10.39% | 0.0031 | [−12.9, −7.9] | 3.1 |
| 44.4° | 3 | −6.92% | 0.0127 | [−10.3, −3.5] | 4.2 |
| *579 scenes: the same ladder, underpowered* | | | | | |
| 0.0° | 3 | −15.8% | 0.021 | [−25.6, −5.9] | 12.3 |
| 5.0° | 3 | −7.6% | 0.0004 | [−8.2, −6.9] | 0.8 |
| 10.0° | 5 | −6.0% | 0.040 | [−11.6, −0.5] | 7.4 |
| 15.1° | 5 | −1.4% | 0.472 | [−6.5, +3.6] | 6.7 |
| 24.9–60.1° | 3 | −0.1 to +2.7% | $\geq$ 0.24 | — | 5.5–9.1 |

## B. Why body facing does not pay

The oracle converts a bearing read off the realised future. Body facing is not that: it is an observation-time estimate of a quantity the model already measures. Three properties of it, on the population the ladder scores, say what is left for it to contribute.

*It is a noisier copy of an existing input.* The box heading estimates the future displacement bearing at 3.9° (+3 s) and 7.1° (+8 s) in median, while SMPL facing reaches 12.3° ($n$=5,824) and 15.1° ($n$=6,206). Box heading is the better estimate at every percentile we measured, the two differ by 10.0° in per-agent median, and both derive from the same returns. We calibrated the SMPL forward axis *against* the box heading before measuring either.

*Most of its accuracy is the trajectory restated.* 95.5% of the scored agents move, and for a mover the pipeline snaps root orientation to the direction of travel whenever the raw estimate deviates by more than 90° (Sec. III). Movers sit at 11.9°, while the 262 stationary agents, the only ones whose facing their velocity does not already imply, sit at 41.2°. The pooled median largely measures the snapping, and the distribution is heavy either way: 5.5° at the first quartile, 23.9° at the third, 45.9° at the ninetieth percentile.

*Where it would be non-redundant, it is worst.* On the one case where velocity carries no directional information (which way a stationary pedestrian steps off), SMPL facing predicts onset direction at roughly twice chance over the 1,705 onset events among 19,939 stationary validation pedestrians: median error 44.2°, 50.6% within 45° (chance 90°, 25%). Informative, consistent with the 41.2° above, and not an artifact of our axis fitting: the box heading estimates the same quantity at 25.2°, and paired on the same events SMPL is 7.3° worse in median (Wilcoxon $p$=2.6×10$^{-43}$). Nor does pose take over where perception degrades: over those events the gap favours the box heading in all ten testable strata ($p < 10^{-5}$ each) and *widens* with range, to +32.9° at 41–75 m, because both derive from the same returns.

Facing is therefore redundant where it is accurate and inaccurate where it would be non-redundant, and neither half of that squeeze is the selection signal the oracle trades on. What survives it is what the content test measures directly: at 25k scenes +0.04%, with the data bounding it to ±1.7%, no more on the benefit side than the ∼1.6% cost of the branch that would carry it.

**Removing the redundant input does not rescue it.** Redundancy says why little is left for facing to add, but it is not by itself a mechanism that blocks conversion. So we tested the implied counterfactual: delete the box-heading channel, retrain all six cells, and repeat the content test, at 579 scenes and eight seeds. Nothing appears, anywhere. The difference in content effect between the two conditions is +2.9 pp on

the 13,006 moving agents, *the wrong sign* for a benefit that redundancy had been masking, and $-0.6$ pp on the stationary ones ($p$=0.65). Seed-level intervals at this scale are wide ($\pm 2$ to $\pm 18$ pp, widest exactly where minADE is largest), so we also average the eight runs per agent and test over agents, which cuts run noise without licensing a claim about retraining: that gives $[+2.4, +3.3]$ pp on movers, $[-1.5, +0.2]$ on the stationary subset, and $-1.2$ pp with $[-2.6, +0.2]$ in the top decile of turning, the stratum where neither velocity nor heading states the future bearing. The two analyses agree to within 0.15 pp on every stratum.

We can also say how much this ablation could ever have shown: deleting the heading channel costs the no-pose baseline only 1.5% on movers, because velocity stays in the input, so there was never much room for facing to fill. What remains unexcluded is the reading the content test's own bound asserts: that pose carries nothing this forecaster can use at any level of redundancy. At $n$=3 this comparison read $-4.1$ pp with all three seeds agreeing in sign: the third multi-percent effect here to dissolve under more seeds.

**A frame-rotation control, not a degraded heading.** The heading the baseline receives is an offboard label too (Sec. IV-E), so it may be better than a deployed tracker's, and facing might pay only where it is worse. Deleting the channel cannot test that: MTR's agent-centric frame, and the intention anchors defined in it, stay aligned to the clean heading. We rotated that frame by a fixed per-agent error $\mathcal{N}(0, \sigma^2)$, which frame, anchors and map inherit, and repeated the content test at 25k scenes and three seeds (Table III). The no-pose baseline loses 5.8% at $\sigma$=15° and 12.6% at 30°, and pose recovers none of it: $-0.14$% ($p$=0.90) and $-0.11$% ($p$=0.93) against shuffled. This does not test a degraded heading, for two reasons. The per-timestep heading channel stays the clean heading expressed in the rotated frame, so the model can read the error we added. And the pose root stays in world coordinates, so the body could not have stated the correction. The control measures the cost of a misaligned frame, and whether facing pays under an onboard-quality heading is untested here. Separately, the walking majority is not drowning out a signal on standing pedestrians: fine-tuning each arm from its clean model for five epochs on standing pedestrians alone leaves pose against shuffled on them at $-1.2$% ($p$=0.28), no better than before ($+1.5$ pp, $p$=0.36). The clean condition reproduces our main cells to within 0.4%.

### C. The cheaper pathway converts a bearing, and still not pose

Early fusion removes the branch tax (Sec. IV-B), which makes it the pathway worth recommending, so we ran the same ladder through it, carrying the oracle in the per-timestep agent channels instead of the branch. At a given level the six channels equal the branch's six inputs bit for bit, so the levels keep their measured-error labels.

At 579 scenes this ladder reads as a dead pathway. No level is significant, *including the uncorrupted one*: $-4.5$% at $\theta$=0 ($p$=0.44), $-1.0$% at 5.0° ($p$=0.53) and $+0.2$% at 10.0° ($p$=0.94), and the per-seed pattern looks like a mechanism

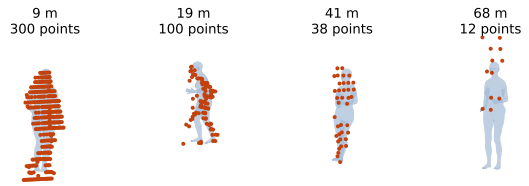



Fig. 3. What the pose estimator has to work with. Raw per-pedestrian LiDAR returns (orange) and the SMPL mesh fitted to them (blue), at four ranges and a common metric scale. Each panel is a frame whose return count equals the median for its range band over the validation segments: 300, 100, 38 and 12 points. Past ≈40 m a pedestrian is a few dozen returns, mostly on head and torso, while the mesh over them still has fully articulated limbs: at that range the articulation is largely the AMASS prior. That applies to the far panels only. Scored against ground-truth keypoints across all ranges, every limb joint carries real skill, and it is the *head* the pipeline emits from its prior (Table V). Sparsity at range is therefore one contributor to the facing error of Sec. V-B, where the pose-versus-heading gap widens to $+32.9$° at 41–75 m, not its whole explanation.

rather than noise. The side branch converts the uncorrupted channel on all three seeds ($-16.5$%, $-19.3$%, $-11.5$%) while early fusion converts it on one ($-14.0$%, against $+0.7$% and $-0.3$%), with the two failing seeds sitting at their no-pose level rather than scattering about it. A difference in *how many seeds move at all* does not look like something a wide interval explains away.

It was. The MDE at $\theta$=0 is 25.5 pp, and we re-ran that level at 25k scenes, where the branch ladder's MDE was 1.4–2.9 pp. Early fusion converts the uncorrupted channel there on *every* seed, by $-31.2$% ($p$=0.0002, 95% CI $[-32.9, -29.5]$), against the branch's $-34.5$% at the same level and denominator, and the corrupted levels follow, $-19.4$% at 5.0° and $-15.5$% at 10.0° ($p \leq 0.001$), each within 0.4 pp of the branch. So the 579 reading was the same power artifact twice over: once in the significance column, and once in a per-seed signature we trusted because it was not a $p$-value. Three runs at 579 scenes cannot tell a pathway that ignores a channel from one that converts it by a third.

This also settles the free-carriage result of Sec. IV-B the other way. "No tax, no benefit" admitted two readings: that early fusion integrates pose at no capacity cost, or that it largely ignores the appended channels, so that there is nothing to cost and nothing to gain. A perfect channel separates them, and at the scale where we can resolve the difference the first reading is correct. Early fusion reads what it is given. Pose through that pathway is therefore a cue that arrives, is used when it carries something, and still buys nothing, which removes the last place the null could have been an artifact of integration rather than a property of the cue.

### D. The complementary cue is barely measured

Gaze precedes locomotion, and head orientation is an established intent cue for vehicle-based pedestrian path prediction [19], [20], so head yaw relative to the torso is a plausible *independent* cue. It is very nearly not recoverable here. In the fitted meshes head-versus-torso yaw has a standard deviation

TABLE V

PER-JOINT RECOVERABILITY AGAINST GROUND-TRUTH KEYPOINTS. MP IS THE ERROR OF ALWAYS PREDICTING THE MEAN POSE (EQUAL TO THE JOINT'S GROUND-TRUTH SPREAD BY CONSTRUCTION), AND $\mathrm{SKILL} = 1 - \mathrm{MPJPE}/\mathrm{MP}$. THE HEAD'S MPJPE BEATS BOTH WRISTS WHILE CARRYING A TENTH OF THEIR SKILL. INTERVALS RESAMPLE PEDESTRIANS ($n$=534), NOT FRAMES, SINCE FRAMES WITHIN A TRACK ARE STRONGLY CORRELATED. EVERY LIMB INTERVAL CLEARS ZERO BY A WIDE MARGIN (LOWEST BOUND $+0.43$). THE HEAD'S IS $[-0.003, +0.089]$, THE ONE INTERVAL THAT CONTAINS IT.

| Joint | MPJPE (mm) | MP (mm) | SKILL |
|---|---|---|---|
| Elbows | 91.2 / 91.5 | 252.9 / 248.9 | $+0.64$ / $+0.63$ |
| Hips | 39.8 | 97.9 | $+0.59$ |
| Wrists | 118.3 / 125.4 | 271.1 / 276.4 | $+0.56$ / $+0.55$ |
| Shoulders | 87.3 / 89.0 | 171.5 / 164.0 | $+0.49$ / $+0.46$ |
| Ankles | 114.0 / 116.6 | 221.6 / 223.8 | $+0.49$ / $+0.48$ |
| Knees | 78.4 / 81.5 | 150.8 / 151.2 | $+0.48$ / $+0.46$ |
| **Head** | **111.6** | **116.9** | $\mathbf{+0.045}$ |

of $3.8°$, while against the ground-truth 3D keypoints released with Waymo Perception [21], [22] ($n$=8,234) the real quantity varies $19.2°$: a fivefold under-dispersion, so the estimator is largely emitting its prior. Consistently, head yaw carries no usable signal for onset direction ($r = -0.03$, n.s.), and the head is the one joint in Table V whose interval includes zero. At 20–40 m the head is a near-spherical handful of points with little geometry to constrain yaw.

### *E. MPJPE is the wrong acceptance metric*

A single mean error cannot distinguish an estimator that tracks a joint imprecisely from one that emits its prior. We therefore score each joint against the strongest trivial competitor, always predicting the mean root-aligned pose: $\mathrm{SKILL}_j = 1 - \mathrm{MPJPE}_j/\mathrm{MP}_j$, where values at or below zero mean the estimator contributes nothing at that joint.

Evaluated on Waymo Perception against ground-truth keypoints (1,789 frames, 534 pedestrians), every limb joint scores $+0.457$ to $+0.639$ (Table V). The head scores $+0.045$, yet its MPJPE (111.6 mm) is *better* than either wrist (118.3/125.4 mm). Its error is flattering not because it is well estimated but because it barely improves on its own mean-pose baseline (116.9 mm), whereas the hips, whose ground-truth spread is *smaller* at 97.9 mm, are predicted at 39.8 mm. Across the thirteen joints MPJPE and skill are essentially uncorrelated (Spearman $\rho = -0.11$, $p$=0.72): the aggregate is not merely a noisy proxy for information content but carries almost none of it, and for the head it actively inverts the ordering. We recommend reporting per-joint skill alongside it whenever a pose estimator is accepted into a downstream stack.

### *F. A general condition*

Two results locate where pose could have helped. At 2 s, minADE is 0.097 m at 25k scenes, and a perfect future-direction oracle through the same branch removes only 1.4 cm of it ($-14.7\%$): relative headroom exists at short horizons, but in absolute terms there is almost nothing for any cue to win. This is a property of the forecaster's own residual, not of constant velocity: single-mode constant-velocity extrapolation [23] gives 0.310 m at 2 s on the same validation split, about three times the model's error, so short-horizon forecasting is *not* CV-saturated (a comparison that favours the model, since minADE takes the best of $K$=6 modes). Saturation here means the baseline's own remaining error is a few percent of the displacement. At long horizons the residual is dominated by destination and intent, which body configuration does not disclose, and stratifying by how badly constant velocity fails still yields no effect.

In general form, and offered as a hypothesis from one modality on one dataset rather than a law: *an auxiliary modality improves a forecast only where the baseline's own residual leaves room in absolute terms, where that residual is not dominated by intent the modality cannot observe, and where the modality carries information about the cue that the model's existing inputs do not already carry.* All three clauses must hold. The third is not a comparison of marginal accuracies, and this paper is the reason to state it that way: Sec. V-A converts a channel less accurate at every percentile than an input the model already has, so being the noisier estimate is not by itself disqualifying. What disqualifies facing is that its error is *shared* with the input it would supplement. It is calibrated against the box heading, snapped to the direction of travel for the 95.5% of agents that move, and recovered from the same returns, so where it is accurate it is redundant and where it would be independent it is worst (Sec. V-B). Sec. IV-E tests the clause from the other side: degrade the input the body duplicates, and the body appears to pay where the sensor resolves it. It also shows the clause has a trap. A cue whose pipeline was fitted to the input it supplements carries that input, and looks informative exactly when the input is degraded. Redundancy is therefore a property of a modality together with the inputs beside it, and an offboard-labelled benchmark sets those inputs at offboard quality. The check this licenses is still cheap and still skipped, and it is still a measurement on the dataset with no training: how much of the target does the candidate cue explain that the existing inputs do not?

## VI. WHAT THIS MEANS IN PRACTICE

For an AV stack, four consequences follow. First, before integrating an auxiliary modality, measure not only its error on the cue it is meant to carry but how much of that cue it explains *beyond the inputs the model already receives*, and take those inputs at the quality a vehicle will have. That costs no training. On offboard labels it predicts our null, and a simulated causal tracker, a few lines in a data loader, shows what the same body is worth on a degraded track, provided the cue's own pipeline never saw the labels. Ask the same of the cue's provenance. Our fitted poses predicted the labelled speed with $R^2$=0.94, and with 0.78 on moving pedestrians seen through fewer than 30 returns, which no such body can, and that one number would have flagged the leak before any training. Second, an oracle ladder is still worth its handful of runs, as a positive control and as an upper bound on the pathway, but size it so

its MDE is below the effect you would act on, and do not read its angular axis as an acceptance threshold: ours converts a channel less accurate than an input the model already has, because an oracle knows which future occurred and no sensor does (Sec. V-A). Third, the capacity cost of an extra branch is real ($\sim$1.6% here, at the best-epoch operating point) and is easily mistaken for a modality effect in either direction. Early fusion avoids it while still reading the channel, as an oracle through that pathway demonstrates, and a shuffled-pose control separates a benefit from a capacity effect either way. Fourth, do not accept a pose estimator on aggregate MPJPE alone: check per-joint skill, because the joints an aggregate flatters may be the ones carrying a prior.

**Limitations.** The largest scale we train is 25k scenes of the 299,220 exported, so stopping there is a compute limit rather than an annotation one. At eight seeds the minimum detectable content effect is $\approx$1.25% at 80% power, and the ladders run at $n$=3 per level, five for track noise (Tables IV and III). We test one forecaster, one dataset and one pose estimator, and the model sees pedestrians and map only. We did not test a degraded heading: the frame-rotation control leaves the clean heading readable, and the default pose arm carries its root in world coordinates (Sec. V-B). We ladder two integration pathways rather than one, but two is not every way a cue can enter a model, and the early-fusion ladder stops at 10.0° (Sec. V-C). The onboard track is simulated: Gaussian detection noise through a constant-velocity Kalman filter rather than a detector's errors. The pre-fit poses are still estimated from points cropped by offboard boxes, the pre-fit control covers the 30 cm level only, and we have not run either through early fusion. The redundancy counterfactual, which deletes the heading channel to see whether facing then pays, is run at 579 scenes rather than at 25k, and its seed-level intervals there are wide enough that the agent-level analysis carries the turn strata. WOMD ships camera data only as quantised tokens, so a camera-based pose or gaze estimate cannot be tested at this scale. On Waymo Perception, where raw imagery exists, an RGB estimator (HMR2.0) reaches 106.2 mm root-aligned MPJPE against LiDAR-HMR's 87.6 mm on the same 347 frames, so the obvious better-pose route is the worse one here. We do not model pose–pose interaction.

## VII. Conclusion

Body pose, recovered from LiDAR, does not improve pedestrian trajectory forecasting on WOMD's offboard tracks. On the offboard labels the null is controlled against a shuffled-pose baseline and holds across scale, representation and integration, and an oracle through the same branch rules out a dead pathway. Under a simulated onboard tracker own pose beats a stranger's by 5.3%, and three quarters of that is the offboard label, carried in by a pose pipeline that fits the gait to the track. Poses that never saw the track gain a non-significant 1.3% overall and 3.5% on the best-resolved bodies, which awaits replication. The results are one mechanism: a cue pays only for what it carries beyond the model's existing inputs, an offboard track leaves the body nothing to add, and a cue fitted to a label carries the label. As a bearing, body facing is a noisier copy of the box heading, and worst, at 41.2°, on the stationary pedestrians it was meant to serve. The oracle ladder shows what the pathway would convert, a direction channel at every error level we trained at 25k scenes, and that what it sells beyond precision is knowledge of which future occurred, which no sensor supplies. The lesson generalises past pose. Before integrating a modality, measure how much of the cue it carries beyond the model's existing inputs, take those inputs at the quality a vehicle will have rather than from offboard labels, check what the cue's own pipeline was fitted to, and make sure any ladder you judge it against can resolve the effect you would act on.

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